Special Session on Machine Learning: How Will Machine Learning Transform Test?

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I. Introduction

This special session will discuss how machine learning can transform test. The first talk will review the key challenges and will argue whether contemporary tools, such as deep learning, could offer any advantages over traditional methods. The second talk will focus on extracting useful information from big test data. The third talk will view adaptive test as running machine learning algorithms in real time on big data.

II. Machine Learning in semiconductor Test: Can Deep Learning Save the Day? (Yiorgos Makris)

While many applications of machine learning in various semiconductor manufacturing and testing tasks have been heavily researched over the last two decades, very few have actually seen the light of day in a real production environment. Recently, the popularity of contemporary artificial intelligence methods, such as deep learning, has reignited the enthusiasm and reinvigorated the discussion regarding the potential of statistical and machine learning-based solutions towards reducing test cost, increasing test quality, improving test floor logistics and providing guidance to both designers and process engineers. In this presentation, we will first review the key challenges in transitioning machine learning-based solutions from a research to a production environment; then, we will discuss the main advantages of contemporary methods, such as deep learning, over traditional machine learning solutions. Ultimately, the intention of this presentation is to caution against considering deep learning a panacea to test (and, thereby, prevent disappointment), as well as to steer attention towards the true operational challenges which need to be addressed in order for any machine learning or statistical solution, traditional or contemporary, to be effective in a real production environment.

III. Small Signals Extraction in Semiconductor Test & Manufacturing: Role of Machine Learning (Amit Nahar)

Semiconductor manufacturing and testing generates huge amount of data. Identifying small signals from this plethora of data and using it to drive product quality improvements are becoming increasingly important, especially for products going into automotive applications. The big data approaches of data management have increased speed, quality and accessibility of the data. This presentation will discuss challenges and applications of harnessing value from the fab manufacturing and test data. A lot of signals are collected in fab manufacturing from fab tools (e.g. pressures, frequencies, temperature etc). The time interval of collecting these signals varies for different tools (i.e. every second, minute or hour). Analyzing this enormous multidimensional time domain data near real-time and finding signals which deviate from normal is challenging. Machine learning algorithms play a key role in finding these deviations and predicting excursions so that appropriate actions can be taken downstream (most likely at test). Another application that will be discussed is finding small signals from test data. Typically hundreds of tests are performed for every product; hence finding small signals for thousands of such products presents a big challenge. Machine learning algorithms running in a big data environment enables extracting these small signals and using them to correlate with customer returns.

IV. Adaptive Test: Machine Learning in Real Time on Big Data (Haralampos-G. Stratigopoulos)

In the standard test approach, the test process is fixed and identical for every circuit and is only occasionally revised, typically every few months, when extensive test data have been collected and confident decisions can be made as to how the test process could be improved to lower the test cost. Adaptive test adheres to the standard test approach, but dynamically adjusts it in real-time to the particularities of each wafer or die instead of having a fixed test process. In this way, significant test time savings and test quality improvements can be achieved. Adjusting the test process may involve adjusting the test limits, the test content, and the test order. The dynamic adjustment may be decided based on available big data: historical (i.e. data from previously tested wafers or lots, customer returns, etc.), near-time (i.e. data from previously tested dies on the same wafer or previously tested wafers in the same lot), and real-time data (i.e. data from already performed tests on the die that is currently under test). The core of an adaptive test flow is the underlying decision-making algorithm. Machine learning algorithms can be very effective in mining available information and identifying meaningful correlations. The challenge is that calculations must be quick such that test application is not delayed, or, if it is delayed, then any such delay is counterbalanced by an average test time reduction per wafer or die or is justified by the improved test quality. This talk will discuss how machine learning can lead to more effective adaptive test flows.

978-1-5386-3774-6/18/$31.00 ©2018 IEEE

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